

by\_normal\_form\_q11 over GF(11)

June 30, 2026

### The equation

The equation of the quartic curve is :

$$X_0^4 + 4X_1^4 + 4X_2^4 + 5X_0^3X_1 + 4X_0^3X_2 + 4X_0X_1^3 + 4X_1^3X_2 + 3X_0X_2^3 + 7X_1X_2^3 + 4X_0^2X_2^2 + 4X_1^2X_2^2 + 10X_0^2X_1X_2 + X_0X_1^2X_2 + X_0X_1X_2^2$$

$$(1, 4, 4, 5, 4, 4, 4, 3, 7, 0, 4, 4, 10, 1, 1)$$

$$X_0^4 + 4X_1^4 + 4X_2^4 + 5X_0^3X_1 + 4X_0^3X_2 + 4X_0X_1^3 + 4X_1^3X_2 + 3X_0X_2^3 + 7X_1X_2^3$$

### General information

|                                       |              |
|---------------------------------------|--------------|
| Number of bitangents                  | 28           |
| Number of points                      | 0            |
| Fullness                              | is full      |
| Number of Kovalevski points           | 21           |
| Bitangent line type $(a_0, a_1, a_2)$ | $(28, 0, 0)$ |
| Number of singular points             | 0            |

### All points on the curve

The quartic has 0 points:

The points on the quartic curve are:

The points by rank are: ( )

The Kovalevski points are:

$$0 : P_{125} = (3, 10, 1) = b_2 \cap b_5 \cap c_{23} \cap c_{35}$$

$$1 : P_{61} = (5, 4, 1) = a_2 \cap a_6 \cap c_{24} \cap c_{46}$$

$$2 : P_{122} = (0, 10, 1) = a_2 \cap a_3 \cap b_2 \cap b_3$$

$$3 : P_{120} = (9, 9, 1) = c_{24} \cap c_{26} \cap c_{34} \cap c_{36}$$

4 :  $P_{28} = (5, 1, 1) = c_{12} \cap c_{13} \cap c_{25} \cap c_{35}$   
 5 :  $P_{109} = (9, 8, 1) = a_2 \cap b_6 \cap c_{15} \cap c_{34}$   
 6 :  $P_{26} = (3, 1, 1) = a_1 \cap a_6 \cap c_{12} \cap c_{26}$   
 7 :  $P_{103} = (3, 8, 1) = c_{14} \cap c_{15} \cap c_{46} \cap c_{56}$   
 8 :  $P_{51} = (6, 3, 1) = c_{16} \cap c_{23} \cap c_{45} \cap d$   
 9 :  $P_{101} = (1, 8, 1) = b_1 \cap b_3 \cap c_{15} \cap c_{35}$   
 10 :  $P_{11} = (8, 1, 0) = a_4 \cap a_6 \cap c_{14} \cap c_{16}$   
 11 :  $P_{94} = (5, 7, 1) = b_1 \cap b_5 \cap c_{16} \cap c_{56}$   
 12 :  $P_{21} = (8, 0, 1) = b_4 \cap b_6 \cap c_{45} \cap c_{56}$   
 13 :  $P_{10} = (7, 1, 0) = b_1 \cap b_6 \cap c_{13} \cap c_{36}$   
 14 :  $P_{78} = (0, 6, 1) = a_3 \cap a_4 \cap c_{23} \cap c_{24}$   
 15 :  $P_{35} = (1, 2, 1) = a_4 \cap b_5 \cap c_{13} \cap c_{26}$   
 16 :  $P_8 = (5, 1, 0) = a_1 \cap b_3 \cap c_{25} \cap c_{46}$   
 17 :  $P_{68} = (1, 5, 1) = a_1 \cap a_5 \cap c_{14} \cap c_{45}$   
 18 :  $P_{128} = (6, 10, 1) = a_5 \cap b_2 \cap c_{25} \cap d$   
 19 :  $P_{31} = (8, 1, 1) = a_5 \cap b_4 \cap c_{12} \cap c_{36}$   
 20 :  $P_1 = (0, 1, 0) = a_3 \cap b_4 \cap c_{34} \cap d$

The incidence structure induced by the Kovalevski points is:

Flags=( 6, 16, 17, 22, 23, 26, 44, 56, 62, 73, 77, 78, 101, 102, 103, 106, 111, 115, 135, 137, 139, 147, 149, 165, 170, 177, 184, 201, 208, 209, 210, 221, 225, 236, 243, 244, 256, 258, 271, 277, 286, 288, 301, 304, 311, 320, 322, 324, 344, 346, 347, 357, 365, 371, 379, 381, 392, 403, 415, 417, 423, 426, 435, 444, 446, 461, 462, 466, 471, 486, 496, 502, 512, 516, 521, 526, 532, 541, 553, 557, 558, 575, 585, 587 )

nb\_rows=28

nb\_cols=21

nb\_flags=84

Incma=

( 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0 )  
 ( 0, 1, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0 )  
 ( 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1 )  
 ( 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 1, 0, 0, 0, 0 )  
 ( 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 0 )  
 ( 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0 )  
 ( 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0 )  
 ( 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0 )  
 ( 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0 )  
 ( 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 1 )  
 ( 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0 )  
 ( 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0 )  
 ( 0, 0, 0, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0 )  
 ( 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 0, 0, 0, 0, 0 )  
 ( 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0 )  
 ( 0, 0, 0, 0, 0, 1, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0 )  
 ( 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0 )  
 ( 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0 )

( 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0 )  
( 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 0, 0 )  
( 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0 )  
( 0, 0, 0, 1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1 )  
( 1, 0, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0 )  
( 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0 )  
( 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0, 0, 0, 1, 0, 0 )  
( 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0 )  
( 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0 )  
( 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 1, 0 )

The Kovalevski points sorted by rank are: ( 1, 8, 10, 11, 21, 26, 28, 31, 35, 51,  
61, 68, 78, 94, 101, 103, 109, 120, 122, 125, 128 )

The points off the curve are:

|                            |                            |                            |
|----------------------------|----------------------------|----------------------------|
| 0 : $P_0 = (1, 0, 0)$      | 31 : $P_{31} = (8, 1, 1)$  | 62 : $P_{62} = (6, 4, 1)$  |
| 1 : $P_1 = (0, 1, 0)$      | 32 : $P_{32} = (9, 1, 1)$  | 63 : $P_{63} = (7, 4, 1)$  |
| 2 : $P_2 = (0, 0, 1)$      | 33 : $P_{33} = (10, 1, 1)$ | 64 : $P_{64} = (8, 4, 1)$  |
| 3 : $P_3 = (1, 1, 1)$      | 34 : $P_{34} = (0, 2, 1)$  | 65 : $P_{65} = (9, 4, 1)$  |
| 4 : $P_4 = (1, 1, 0)$      | 35 : $P_{35} = (1, 2, 1)$  | 66 : $P_{66} = (10, 4, 1)$ |
| 5 : $P_5 = (2, 1, 0)$      | 36 : $P_{36} = (2, 2, 1)$  | 67 : $P_{67} = (0, 5, 1)$  |
| 6 : $P_6 = (3, 1, 0)$      | 37 : $P_{37} = (3, 2, 1)$  | 68 : $P_{68} = (1, 5, 1)$  |
| 7 : $P_7 = (4, 1, 0)$      | 38 : $P_{38} = (4, 2, 1)$  | 69 : $P_{69} = (2, 5, 1)$  |
| 8 : $P_8 = (5, 1, 0)$      | 39 : $P_{39} = (5, 2, 1)$  | 70 : $P_{70} = (3, 5, 1)$  |
| 9 : $P_9 = (6, 1, 0)$      | 40 : $P_{40} = (6, 2, 1)$  | 71 : $P_{71} = (4, 5, 1)$  |
| 10 : $P_{10} = (7, 1, 0)$  | 41 : $P_{41} = (7, 2, 1)$  | 72 : $P_{72} = (5, 5, 1)$  |
| 11 : $P_{11} = (8, 1, 0)$  | 42 : $P_{42} = (8, 2, 1)$  | 73 : $P_{73} = (6, 5, 1)$  |
| 12 : $P_{12} = (9, 1, 0)$  | 43 : $P_{43} = (9, 2, 1)$  | 74 : $P_{74} = (7, 5, 1)$  |
| 13 : $P_{13} = (10, 1, 0)$ | 44 : $P_{44} = (10, 2, 1)$ | 75 : $P_{75} = (8, 5, 1)$  |
| 14 : $P_{14} = (1, 0, 1)$  | 45 : $P_{45} = (0, 3, 1)$  | 76 : $P_{76} = (9, 5, 1)$  |
| 15 : $P_{15} = (2, 0, 1)$  | 46 : $P_{46} = (1, 3, 1)$  | 77 : $P_{77} = (10, 5, 1)$ |
| 16 : $P_{16} = (3, 0, 1)$  | 47 : $P_{47} = (2, 3, 1)$  | 78 : $P_{78} = (0, 6, 1)$  |
| 17 : $P_{17} = (4, 0, 1)$  | 48 : $P_{48} = (3, 3, 1)$  | 79 : $P_{79} = (1, 6, 1)$  |
| 18 : $P_{18} = (5, 0, 1)$  | 49 : $P_{49} = (4, 3, 1)$  | 80 : $P_{80} = (2, 6, 1)$  |
| 19 : $P_{19} = (6, 0, 1)$  | 50 : $P_{50} = (5, 3, 1)$  | 81 : $P_{81} = (3, 6, 1)$  |
| 20 : $P_{20} = (7, 0, 1)$  | 51 : $P_{51} = (6, 3, 1)$  | 82 : $P_{82} = (4, 6, 1)$  |
| 21 : $P_{21} = (8, 0, 1)$  | 52 : $P_{52} = (7, 3, 1)$  | 83 : $P_{83} = (5, 6, 1)$  |
| 22 : $P_{22} = (9, 0, 1)$  | 53 : $P_{53} = (8, 3, 1)$  | 84 : $P_{84} = (6, 6, 1)$  |
| 23 : $P_{23} = (10, 0, 1)$ | 54 : $P_{54} = (9, 3, 1)$  | 85 : $P_{85} = (7, 6, 1)$  |
| 24 : $P_{24} = (0, 1, 1)$  | 55 : $P_{55} = (10, 3, 1)$ | 86 : $P_{86} = (8, 6, 1)$  |
| 25 : $P_{25} = (2, 1, 1)$  | 56 : $P_{56} = (0, 4, 1)$  | 87 : $P_{87} = (9, 6, 1)$  |
| 26 : $P_{26} = (3, 1, 1)$  | 57 : $P_{57} = (1, 4, 1)$  | 88 : $P_{88} = (10, 6, 1)$ |
| 27 : $P_{27} = (4, 1, 1)$  | 58 : $P_{58} = (2, 4, 1)$  | 89 : $P_{89} = (0, 7, 1)$  |
| 28 : $P_{28} = (5, 1, 1)$  | 59 : $P_{59} = (3, 4, 1)$  | 90 : $P_{90} = (1, 7, 1)$  |
| 29 : $P_{29} = (6, 1, 1)$  | 60 : $P_{60} = (4, 4, 1)$  | 91 : $P_{91} = (2, 7, 1)$  |
| 30 : $P_{30} = (7, 1, 1)$  | 61 : $P_{61} = (5, 4, 1)$  | 92 : $P_{92} = (3, 7, 1)$  |

|                             |                              |                               |
|-----------------------------|------------------------------|-------------------------------|
| 93 : $P_{93} = (4, 7, 1)$   | 107 : $P_{107} = (7, 8, 1)$  | 121 : $P_{121} = (10, 9, 1)$  |
| 94 : $P_{94} = (5, 7, 1)$   | 108 : $P_{108} = (8, 8, 1)$  | 122 : $P_{122} = (0, 10, 1)$  |
| 95 : $P_{95} = (6, 7, 1)$   | 109 : $P_{109} = (9, 8, 1)$  | 123 : $P_{123} = (1, 10, 1)$  |
| 96 : $P_{96} = (7, 7, 1)$   | 110 : $P_{110} = (10, 8, 1)$ | 124 : $P_{124} = (2, 10, 1)$  |
| 97 : $P_{97} = (8, 7, 1)$   | 111 : $P_{111} = (0, 9, 1)$  | 125 : $P_{125} = (3, 10, 1)$  |
| 98 : $P_{98} = (9, 7, 1)$   | 112 : $P_{112} = (1, 9, 1)$  | 126 : $P_{126} = (4, 10, 1)$  |
| 99 : $P_{99} = (10, 7, 1)$  | 113 : $P_{113} = (2, 9, 1)$  | 127 : $P_{127} = (5, 10, 1)$  |
| 100 : $P_{100} = (0, 8, 1)$ | 114 : $P_{114} = (3, 9, 1)$  | 128 : $P_{128} = (6, 10, 1)$  |
| 101 : $P_{101} = (1, 8, 1)$ | 115 : $P_{115} = (4, 9, 1)$  | 129 : $P_{129} = (7, 10, 1)$  |
| 102 : $P_{102} = (2, 8, 1)$ | 116 : $P_{116} = (5, 9, 1)$  | 130 : $P_{130} = (8, 10, 1)$  |
| 103 : $P_{103} = (3, 8, 1)$ | 117 : $P_{117} = (6, 9, 1)$  | 131 : $P_{131} = (9, 10, 1)$  |
| 104 : $P_{104} = (4, 8, 1)$ | 118 : $P_{118} = (7, 9, 1)$  | 132 : $P_{132} = (10, 10, 1)$ |
| 105 : $P_{105} = (5, 8, 1)$ | 119 : $P_{119} = (8, 9, 1)$  |                               |
| 106 : $P_{106} = (6, 8, 1)$ | 120 : $P_{120} = (9, 9, 1)$  |                               |

( 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128, 129, 130, 131, 132 )

## Duals of Bitangents

dual of bitangents:

- 0 :  $P_{51} = (6, 3, 1)$
- 1 :  $P_{33} = (10, 1, 1)$
- 2 :  $P_0 = (1, 0, 0)$
- 3 :  $P_{114} = (3, 9, 1)$
- 4 :  $P_{103} = (3, 8, 1)$
- 5 :  $P_{76} = (9, 5, 1)$
- 6 :  $P_6 = (3, 1, 0)$
- 7 :  $P_{24} = (0, 1, 1)$
- 8 :  $P_{25} = (2, 1, 1)$
- 9 :  $P_{17} = (4, 0, 1)$
- 10 :  $P_{87} = (9, 6, 1)$
- 11 :  $P_{71} = (4, 5, 1)$
- 12 :  $P_{122} = (0, 10, 1)$
- 13 :  $P_{40} = (6, 2, 1)$
- 14 :  $P_{80} = (2, 6, 1)$
- 15 :  $P_{56} = (0, 4, 1)$
- 16 :  $P_{42} = (8, 2, 1)$
- 17 :  $P_{121} = (10, 9, 1)$
- 18 :  $P_{119} = (8, 9, 1)$

$$\begin{aligned}
19 : P_5 &= (2, 1, 0) \\
20 : P_{58} &= (2, 4, 1) \\
21 : P_{19} &= (6, 0, 1) \\
22 : P_{53} &= (8, 3, 1) \\
23 : P_{99} &= (10, 7, 1) \\
24 : P_{126} &= (4, 10, 1) \\
25 : P_{92} &= (3, 7, 1) \\
26 : P_{104} &= (4, 8, 1) \\
27 : P_{22} &= (9, 0, 1)
\end{aligned}$$

Dimension of the vanishing ideal is 0

The lines and their points of contact are:

$$\begin{aligned}
a_1 &= \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 8 \end{bmatrix}_{68} \\
a_2 &= \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 10 \end{bmatrix}_{22} \\
a_3 &= \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{132} \\
a_4 &= \begin{bmatrix} 1 & 0 & 8 \\ 0 & 1 & 2 \end{bmatrix}_{98} \\
a_5 &= \begin{bmatrix} 1 & 0 & 8 \\ 0 & 1 & 3 \end{bmatrix}_{99} \\
a_6 &= \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 6 \end{bmatrix}_{30} \\
b_1 &= \begin{bmatrix} 1 & 8 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{107} \\
b_2 &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 10 \end{bmatrix}_{10} \\
b_3 &= \begin{bmatrix} 1 & 0 & 9 \\ 0 & 1 & 10 \end{bmatrix}_{118} \\
b_4 &= \begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & 0 \end{bmatrix}_{84} \\
b_5 &= \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 5 \end{bmatrix}_{29} \\
b_6 &= \begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & 6 \end{bmatrix}_{90} \\
c_{12} &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}_1 \\
c_{13} &= \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 9 \end{bmatrix}_{69} \\
c_{14} &= \begin{bmatrix} 1 & 0 & 9 \\ 0 & 1 & 5 \end{bmatrix}_{113}
\end{aligned}$$

$$\begin{aligned}
c_{15} &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 7 \end{bmatrix}_7 \\
c_{16} &= \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 9 \end{bmatrix}_{45} \\
c_{23} &= \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix}_{14} \\
c_{24} &= \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 2 \end{bmatrix}_{38} \\
c_{25} &= \begin{bmatrix} 1 & 9 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{119} \\
c_{26} &= \begin{bmatrix} 1 & 0 & 9 \\ 0 & 1 & 7 \end{bmatrix}_{115} \\
c_{34} &= \begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 0 \end{bmatrix}_{60} \\
c_{35} &= \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 8 \end{bmatrix}_{44} \\
c_{36} &= \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 4 \end{bmatrix}_{16} \\
c_{45} &= \begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & 1 \end{bmatrix}_{85} \\
c_{46} &= \begin{bmatrix} 1 & 0 & 8 \\ 0 & 1 & 4 \end{bmatrix}_{100} \\
c_{56} &= \begin{bmatrix} 1 & 0 & 7 \\ 0 & 1 & 3 \end{bmatrix}_{87} \\
d &= \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \end{bmatrix}_{24}
\end{aligned}$$

Rank of lines: ( 68, 22, 132, 98, 99, 30, 107, 10, 118, 84, 29, 90, 1, 69, 113, 7, 45, 14, 38, 119, 115, 60, 44, 16, 85, 100, 87, 24 )

Line type:  $0^{28}$

|    |   |                                                                                                            |
|----|---|------------------------------------------------------------------------------------------------------------|
| 28 | 0 | 1, 2, 3, 4, 5, 6, 7, 8, 9, 10,<br>11, 12, 13, 14, 15, 16, 17, 18, 19, 20,<br>21, 22, 23, 24, 25, 26, 27, 0 |
|----|---|------------------------------------------------------------------------------------------------------------|

point types:

point types for points off the curve:  $4^{21}$ ,  $3^{84}$ ,  $0^{28}$

|    |   |                                                                                                                                                                                                                                                                                                                                                                                          |
|----|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 21 | 4 | 125, 61, 122, 120, 28, 109, 26, 103, 51, 101,<br>11, 94, 21, 10, 78, 35, 8, 68, 128, 31,<br>1                                                                                                                                                                                                                                                                                            |
| 84 | 3 | 30, 14, 29, 116, 55, 27, 6, 121, 123, 59,<br>49, 47, 2, 119, 113, 22, 91, 45, 95, 97,<br>100, 24, 12, 52, 53, 60, 111, 56, 44, 117,<br>43, 4, 126, 42, 84, 20, 40, 80, 19, 62,<br>18, 37, 130, 73, 17, 131, 75, 76, 77, 38,<br>129, 70, 69, 82, 34, 33, 85, 86, 87, 88,<br>89, 67, 0, 92, 65, 32, 64, 96, 63, 98,<br>127, 3, 132, 102, 25, 104, 105, 106, 107, 108,<br>54, 110, 124, 112 |
| 28 | 0 | 58, 46, 15, 115, 74, 99, 83, 71, 114, 118,<br>16, 36, 90, 50, 48, 41, 79, 72, 23, 5,<br>57, 13, 7, 66, 93, 9, 39, 81                                                                                                                                                                                                                                                                     |

Lines on points off the curve:

Off point 0 =  $P_0 = (1, 0, 0)$  lies on 3 bisecants : { 7, 12, 15 }  
Off point 1 =  $P_1 = (0, 1, 0)$  lies on 4 bisecants : { 2, 9, 21, 27 }  
Off point 2 =  $P_2 = (0, 0, 1)$  lies on 3 bisecants : { 2, 6, 19 }  
Off point 3 =  $P_3 = (1, 1, 1)$  lies on 3 bisecants : { 12, 16, 25 }  
Off point 4 =  $P_4 = (1, 1, 0)$  lies on 3 bisecants : { 1, 4, 22 }  
Off point 5 =  $P_5 = (2, 1, 0)$  lies on 0 bisecants : { }  
Off point 6 =  $P_6 = (3, 1, 0)$  lies on 3 bisecants : { 10, 18, 24 }  
Off point 7 =  $P_7 = (4, 1, 0)$  lies on 0 bisecants : { }  
Off point 8 =  $P_8 = (5, 1, 0)$  lies on 4 bisecants : { 0, 8, 19, 25 }  
Off point 9 =  $P_9 = (6, 1, 0)$  lies on 0 bisecants : { }  
Off point 10 =  $P_{10} = (7, 1, 0)$  lies on 4 bisecants : { 6, 11, 13, 23 }  
Off point 11 =  $P_{11} = (8, 1, 0)$  lies on 4 bisecants : { 3, 5, 14, 16 }  
Off point 12 =  $P_{12} = (9, 1, 0)$  lies on 3 bisecants : { 17, 20, 26 }  
Off point 13 =  $P_{13} = (10, 1, 0)$  lies on 0 bisecants : { }  
Off point 14 =  $P_{14} = (1, 0, 1)$  lies on 3 bisecants : { 1, 17, 23 }  
Off point 15 =  $P_{15} = (2, 0, 1)$  lies on 0 bisecants : { }  
Off point 16 =  $P_{16} = (3, 0, 1)$  lies on 0 bisecants : { }  
Off point 17 =  $P_{17} = (4, 0, 1)$  lies on 3 bisecants : { 16, 18, 22 }  
Off point 18 =  $P_{18} = (5, 0, 1)$  lies on 3 bisecants : { 8, 14, 20 }  
Off point 19 =  $P_{19} = (6, 0, 1)$  lies on 3 bisecants : { 5, 10, 27 }  
Off point 20 =  $P_{20} = (7, 0, 1)$  lies on 3 bisecants : { 3, 4, 25 }  
Off point 21 =  $P_{21} = (8, 0, 1)$  lies on 4 bisecants : { 9, 11, 24, 26 }  
Off point 22 =  $P_{22} = (9, 0, 1)$  lies on 3 bisecants : { 0, 13, 21 }  
Off point 23 =  $P_{23} = (10, 0, 1)$  lies on 0 bisecants : { }  
Off point 24 =  $P_{24} = (0, 1, 1)$  lies on 3 bisecants : { 2, 12, 24 }  
Off point 25 =  $P_{25} = (2, 1, 1)$  lies on 3 bisecants : { 1, 12, 14 }  
Off point 26 =  $P_{26} = (3, 1, 1)$  lies on 4 bisecants : { 0, 5, 12, 20 }

Off point 27 =  $P_{27} = (4, 1, 1)$  lies on 3 bisecants :  $\{ 3, 11, 12 \}$   
 Off point 28 =  $P_{28} = (5, 1, 1)$  lies on 4 bisecants :  $\{ 12, 13, 19, 22 \}$   
 Off point 29 =  $P_{29} = (6, 1, 1)$  lies on 3 bisecants :  $\{ 12, 26, 27 \}$   
 Off point 30 =  $P_{30} = (7, 1, 1)$  lies on 3 bisecants :  $\{ 6, 12, 18 \}$   
 Off point 31 =  $P_{31} = (8, 1, 1)$  lies on 4 bisecants :  $\{ 4, 9, 12, 23 \}$   
 Off point 32 =  $P_{32} = (9, 1, 1)$  lies on 3 bisecants :  $\{ 10, 12, 21 \}$   
 Off point 33 =  $P_{33} = (10, 1, 1)$  lies on 3 bisecants :  $\{ 8, 12, 17 \}$   
 Off point 34 =  $P_{34} = (0, 2, 1)$  lies on 3 bisecants :  $\{ 2, 5, 11 \}$   
 Off point 35 =  $P_{35} = (1, 2, 1)$  lies on 4 bisecants :  $\{ 3, 10, 13, 20 \}$   
 Off point 36 =  $P_{36} = (2, 2, 1)$  lies on 0 bisecants :  $\{ \}$   
 Off point 37 =  $P_{37} = (3, 2, 1)$  lies on 3 bisecants :  $\{ 1, 6, 24 \}$   
 Off point 38 =  $P_{38} = (4, 2, 1)$  lies on 3 bisecants :  $\{ 8, 23, 26 \}$   
 Off point 39 =  $P_{39} = (5, 2, 1)$  lies on 0 bisecants :  $\{ \}$   
 Off point 40 =  $P_{40} = (6, 2, 1)$  lies on 3 bisecants :  $\{ 22, 25, 27 \}$   
 Off point 41 =  $P_{41} = (7, 2, 1)$  lies on 0 bisecants :  $\{ \}$   
 Off point 42 =  $P_{42} = (8, 2, 1)$  lies on 3 bisecants :  $\{ 0, 9, 17 \}$   
 Off point 43 =  $P_{43} = (9, 2, 1)$  lies on 3 bisecants :  $\{ 4, 16, 21 \}$   
 Off point 44 =  $P_{44} = (10, 2, 1)$  lies on 3 bisecants :  $\{ 14, 18, 19 \}$   
 Off point 45 =  $P_{45} = (0, 3, 1)$  lies on 3 bisecants :  $\{ 2, 23, 25 \}$   
 Off point 46 =  $P_{46} = (1, 3, 1)$  lies on 0 bisecants :  $\{ \}$   
 Off point 47 =  $P_{47} = (2, 3, 1)$  lies on 3 bisecants :  $\{ 0, 18, 26 \}$   
 Off point 48 =  $P_{48} = (3, 3, 1)$  lies on 0 bisecants :  $\{ \}$   
 Off point 49 =  $P_{49} = (4, 3, 1)$  lies on 3 bisecants :  $\{ 1, 10, 19 \}$   
 Off point 50 =  $P_{50} = (5, 3, 1)$  lies on 0 bisecants :  $\{ \}$   
 Off point 51 =  $P_{51} = (6, 3, 1)$  lies on 4 bisecants :  $\{ 16, 17, 24, 27 \}$   
 Off point 52 =  $P_{52} = (7, 3, 1)$  lies on 3 bisecants :  $\{ 11, 14, 22 \}$   
 Off point 53 =  $P_{53} = (8, 3, 1)$  lies on 3 bisecants :  $\{ 5, 9, 13 \}$   
 Off point 54 =  $P_{54} = (9, 3, 1)$  lies on 3 bisecants :  $\{ 3, 8, 21 \}$   
 Off point 55 =  $P_{55} = (10, 3, 1)$  lies on 3 bisecants :  $\{ 4, 6, 20 \}$   
 Off point 56 =  $P_{56} = (0, 4, 1)$  lies on 3 bisecants :  $\{ 2, 4, 26 \}$   
 Off point 57 =  $P_{57} = (1, 4, 1)$  lies on 0 bisecants :  $\{ \}$   
 Off point 58 =  $P_{58} = (2, 4, 1)$  lies on 0 bisecants :  $\{ \}$   
 Off point 59 =  $P_{59} = (3, 4, 1)$  lies on 3 bisecants :  $\{ 8, 11, 16 \}$   
 Off point 60 =  $P_{60} = (4, 4, 1)$  lies on 3 bisecants :  $\{ 13, 14, 17 \}$   
 Off point 61 =  $P_{61} = (5, 4, 1)$  lies on 4 bisecants :  $\{ 1, 5, 18, 25 \}$   
 Off point 62 =  $P_{62} = (6, 4, 1)$  lies on 3 bisecants :  $\{ 3, 6, 27 \}$   
 Off point 63 =  $P_{63} = (7, 4, 1)$  lies on 3 bisecants :  $\{ 0, 10, 23 \}$   
 Off point 64 =  $P_{64} = (8, 4, 1)$  lies on 3 bisecants :  $\{ 9, 20, 22 \}$   
 Off point 65 =  $P_{65} = (9, 4, 1)$  lies on 3 bisecants :  $\{ 19, 21, 24 \}$   
 Off point 66 =  $P_{66} = (10, 4, 1)$  lies on 0 bisecants :  $\{ \}$   
 Off point 67 =  $P_{67} = (0, 5, 1)$  lies on 3 bisecants :  $\{ 2, 13, 16 \}$   
 Off point 68 =  $P_{68} = (1, 5, 1)$  lies on 4 bisecants :  $\{ 0, 4, 14, 24 \}$   
 Off point 69 =  $P_{69} = (2, 5, 1)$  lies on 3 bisecants :  $\{ 5, 6, 17 \}$   
 Off point 70 =  $P_{70} = (3, 5, 1)$  lies on 3 bisecants :  $\{ 3, 19, 23 \}$   
 Off point 71 =  $P_{71} = (4, 5, 1)$  lies on 0 bisecants :  $\{ \}$   
 Off point 72 =  $P_{72} = (5, 5, 1)$  lies on 0 bisecants :  $\{ \}$

Off point 73 =  $P_{73} = (6, 5, 1)$  lies on 3 bisecants : { 1, 20, 27 }  
 Off point 74 =  $P_{74} = (7, 5, 1)$  lies on 0 bisecants : { }  
 Off point 75 =  $P_{75} = (8, 5, 1)$  lies on 3 bisecants : { 8, 9, 18 }  
 Off point 76 =  $P_{76} = (9, 5, 1)$  lies on 3 bisecants : { 21, 22, 26 }  
 Off point 77 =  $P_{77} = (10, 5, 1)$  lies on 3 bisecants : { 10, 11, 25 }  
 Off point 78 =  $P_{78} = (0, 6, 1)$  lies on 4 bisecants : { 2, 3, 17, 18 }  
 Off point 79 =  $P_{79} = (1, 6, 1)$  lies on 0 bisecants : { }  
 Off point 80 =  $P_{80} = (2, 6, 1)$  lies on 3 bisecants : { 4, 8, 10 }  
 Off point 81 =  $P_{81} = (3, 6, 1)$  lies on 0 bisecants : { }  
 Off point 82 =  $P_{82} = (4, 6, 1)$  lies on 3 bisecants : { 20, 24, 25 }  
 Off point 83 =  $P_{83} = (5, 6, 1)$  lies on 0 bisecants : { }  
 Off point 84 =  $P_{84} = (6, 6, 1)$  lies on 3 bisecants : { 0, 11, 27 }  
 Off point 85 =  $P_{85} = (7, 6, 1)$  lies on 3 bisecants : { 1, 13, 26 }  
 Off point 86 =  $P_{86} = (8, 6, 1)$  lies on 3 bisecants : { 9, 16, 19 }  
 Off point 87 =  $P_{87} = (9, 6, 1)$  lies on 3 bisecants : { 6, 14, 21 }  
 Off point 88 =  $P_{88} = (10, 6, 1)$  lies on 3 bisecants : { 5, 22, 23 }  
 Off point 89 =  $P_{89} = (0, 7, 1)$  lies on 3 bisecants : { 0, 2, 22 }  
 Off point 90 =  $P_{90} = (1, 7, 1)$  lies on 0 bisecants : { }  
 Off point 91 =  $P_{91} = (2, 7, 1)$  lies on 3 bisecants : { 11, 19, 20 }  
 Off point 92 =  $P_{92} = (3, 7, 1)$  lies on 3 bisecants : { 4, 13, 18 }  
 Off point 93 =  $P_{93} = (4, 7, 1)$  lies on 0 bisecants : { }  
 Off point 94 =  $P_{94} = (5, 7, 1)$  lies on 4 bisecants : { 6, 10, 16, 26 }  
 Off point 95 =  $P_{95} = (6, 7, 1)$  lies on 3 bisecants : { 14, 23, 27 }  
 Off point 96 =  $P_{96} = (7, 7, 1)$  lies on 3 bisecants : { 5, 8, 24 }  
 Off point 97 =  $P_{97} = (8, 7, 1)$  lies on 3 bisecants : { 1, 3, 9 }  
 Off point 98 =  $P_{98} = (9, 7, 1)$  lies on 3 bisecants : { 17, 21, 25 }  
 Off point 99 =  $P_{99} = (10, 7, 1)$  lies on 0 bisecants : { }  
 Off point 100 =  $P_{100} = (0, 8, 1)$  lies on 3 bisecants : { 2, 15, 20 }  
 Off point 101 =  $P_{101} = (1, 8, 1)$  lies on 4 bisecants : { 6, 8, 15, 22 }  
 Off point 102 =  $P_{102} = (2, 8, 1)$  lies on 3 bisecants : { 15, 16, 23 }  
 Off point 103 =  $P_{103} = (3, 8, 1)$  lies on 4 bisecants : { 14, 15, 25, 26 }  
 Off point 104 =  $P_{104} = (4, 8, 1)$  lies on 3 bisecants : { 4, 5, 15 }  
 Off point 105 =  $P_{105} = (5, 8, 1)$  lies on 3 bisecants : { 0, 3, 15 }  
 Off point 106 =  $P_{106} = (6, 8, 1)$  lies on 3 bisecants : { 15, 18, 27 }  
 Off point 107 =  $P_{107} = (7, 8, 1)$  lies on 3 bisecants : { 15, 17, 19 }  
 Off point 108 =  $P_{108} = (8, 8, 1)$  lies on 3 bisecants : { 9, 10, 15 }  
 Off point 109 =  $P_{109} = (9, 8, 1)$  lies on 4 bisecants : { 1, 11, 15, 21 }  
 Off point 110 =  $P_{110} = (10, 8, 1)$  lies on 3 bisecants : { 13, 15, 24 }  
 Off point 111 =  $P_{111} = (0, 9, 1)$  lies on 3 bisecants : { 2, 10, 14 }  
 Off point 112 =  $P_{112} = (1, 9, 1)$  lies on 3 bisecants : { 5, 19, 26 }  
 Off point 113 =  $P_{113} = (2, 9, 1)$  lies on 3 bisecants : { 3, 22, 24 }  
 Off point 114 =  $P_{114} = (3, 9, 1)$  lies on 0 bisecants : { }  
 Off point 115 =  $P_{115} = (4, 9, 1)$  lies on 0 bisecants : { }  
 Off point 116 =  $P_{116} = (5, 9, 1)$  lies on 3 bisecants : { 4, 11, 17 }  
 Off point 117 =  $P_{117} = (6, 9, 1)$  lies on 3 bisecants : { 8, 13, 27 }  
 Off point 118 =  $P_{118} = (7, 9, 1)$  lies on 0 bisecants : { }

Off point 119 =  $P_{119} = (8, 9, 1)$  lies on 3 bisecants :  $\{ 6, 9, 25 \}$   
 Off point 120 =  $P_{120} = (9, 9, 1)$  lies on 4 bisecants :  $\{ 18, 20, 21, 23 \}$   
 Off point 121 =  $P_{121} = (10, 9, 1)$  lies on 3 bisecants :  $\{ 0, 1, 16 \}$   
 Off point 122 =  $P_{122} = (0, 10, 1)$  lies on 4 bisecants :  $\{ 1, 2, 7, 8 \}$   
 Off point 123 =  $P_{123} = (1, 10, 1)$  lies on 3 bisecants :  $\{ 7, 11, 18 \}$   
 Off point 124 =  $P_{124} = (2, 10, 1)$  lies on 3 bisecants :  $\{ 7, 13, 25 \}$   
 Off point 125 =  $P_{125} = (3, 10, 1)$  lies on 4 bisecants :  $\{ 7, 10, 17, 22 \}$   
 Off point 126 =  $P_{126} = (4, 10, 1)$  lies on 3 bisecants :  $\{ 0, 6, 7 \}$   
 Off point 127 =  $P_{127} = (5, 10, 1)$  lies on 3 bisecants :  $\{ 7, 23, 24 \}$   
 Off point 128 =  $P_{128} = (6, 10, 1)$  lies on 4 bisecants :  $\{ 4, 7, 19, 27 \}$   
 Off point 129 =  $P_{129} = (7, 10, 1)$  lies on 3 bisecants :  $\{ 7, 16, 20 \}$   
 Off point 130 =  $P_{130} = (8, 10, 1)$  lies on 3 bisecants :  $\{ 7, 9, 14 \}$   
 Off point 131 =  $P_{131} = (9, 10, 1)$  lies on 3 bisecants :  $\{ 5, 7, 21 \}$   
 Off point 132 =  $P_{132} = (10, 10, 1)$  lies on 3 bisecants :  $\{ 3, 7, 26 \}$

## The gradient

The gradient of the quartic curve is :

$$4X_0^3 + 4X_1^3 + 3X_2^3 + 4X_0^2X_1 + X_0^2X_2 + X_1^2X_2 + 8X_0X_2^2 + X_1X_2^2 + 9X_0X_1X_2$$

$$(4, 4, 3, 4, 1, 0, 1, 8, 1, 9)$$

$$5X_0^3 + 5X_1^3 + 7X_2^3 + 10X_0^2X_2 + X_0X_1^2 + X_1^2X_2 + X_0X_2^2 + 8X_1X_2^2 + 2X_0X_1X_2$$

$$(5, 5, 7, 0, 10, 1, 1, 1, 8, 2)$$

$$4X_0^3 + 4X_1^3 + 5X_2^3 + 10X_0^2X_1 + 8X_0^2X_2 + X_0X_1^2 + 8X_1^2X_2 + 9X_0X_2^2 + 10X_1X_2^2 + 2X_0X_1X_2$$

$$(4, 4, 5, 10, 8, 1, 8, 9, 10, 2)$$